

Adaptive Multi-Sensor Predictive Smart Irrigation System Based on Soil Behaviour Modelling

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Abstract—The efficient management of water is an important problem in modern agriculture due to water scarcity and inefficient traditional irrigation systems. This paper presents an adaptive multi-sensor predictive smart irrigation system based on soil behaviour modelling. Soil moisture, temperature, and humidity sensors are used in conjunction with an ESP32 microcontroller to sense environmental conditions in real time. These conditions are sent to the cloud platform for storage, analysis, and visualization. Unlike conventional irrigation systems based only on threshold conditions, this system uses predictive modelling to analyze trends in soil moisture. A drop in moisture rate is used to predict future conditions of the soil, ensuring efficient water usage through automatic operation of a water pump via a relay mechanism. The experimental results show efficient water usage without waste, maintaining optimal moisture in the soil. This proposed system is efficient, cost-effective, and can be used in precision agriculture.

Keywords—Smart irrigation system, Internet of Things (IoT), ESP32, Soil moisture, Predictive modelling, Precision agriculture, Cloud-based monitoring, Soil behaviour modelling.

I. INTRODUCTION

A. Background

Agriculture is considered to be one of the key areas of human life, and water management is an essential factor in enhancing crop yield. Conventional irrigation systems involve manual operation and are based on fixed schedules. These may cause over-irrigation or under-irrigation. With advancements in technology, intelligent irrigation systems have come into existence.

B. Problem Statement

The conventional irrigation system lacks real-time monitoring and is not adaptable to changing conditions. Most conventional irrigation systems are based on fixed threshold conditions and do not take into consideration changing conditions such as temperature, humidity, and soil conditions. These changing conditions may cause inefficient water usage.

C. Motivation

The scarcity of water resources is an issue of major concern. In addition, sustainable agriculture is also an issue of concern. These are the key motivations behind developing an intelligent irrigation system. With the integration of sensor devices and cloud computing, it is possible to monitor current conditions and also predict future conditions.

D. Objectives

The objectives of this project are:

- Develop an intelligent irrigation system using multiple sensors.
- Monitor soil conditions such as moisture, temperature, and humidity.
- Develop a predictive model based on soil conditions.
- Automate irrigation using an ESP32 microcontroller.
- Reduce water wastage.
- Predictive pest management: Identify high-risk weather windows and alert users before an outbreak.

- Data-driven resource allocation: Automate scheduling of fertilizers and harvesting based on real-time data.
- Low-cost remote sensing: Eliminate the need for expensive on-field hardware by utilizing satellite-derived telemetry.

II. LITERATURE REVIEW

With the rapid advancement in smart agriculture technologies, IoT-based irrigation systems have been proposed to efficiently utilize water and increase crop productivity. The concept of IoT-based irrigation systems has been explored in many research studies, in which soil moisture sensors have been used to monitor and control irrigation systems. It has been shown that sensor-based irrigation systems can efficiently reduce water wastage by accurately measuring soil moisture and automatically controlling irrigation systems. Moreover, sensor-based irrigation systems can efficiently reduce water wastage by accurately measuring soil moisture and automatically controlling irrigation systems.

Further advancements in IoT-based irrigation systems include the integration of other environmental sensors, i.e., temperature and humidity sensors. The integration of these sensors can provide a better understanding of soil and environmental conditions, which can help in increasing crop productivity. This can be achieved by optimizing irrigation systems and increasing crop productivity.

Another advancement in IoT-based irrigation systems is the use of cloud-based IoT architecture. The cloud-based IoT architecture has been widely used in smart irrigation systems. The cloud-based IoT architecture can efficiently provide real-time data transmission, storage, and visualization. The cloud-based IoT architecture has been widely used in smart irrigation systems. It has been shown in many research studies that cloud-based IoT-based irrigation systems can efficiently provide real-time data transmission, storage, and visualization. Moreover, cloud-based IoT-based irrigation systems can efficiently provide real-time data transmission, storage, and visualization.

Furthermore, many researchers have used weather forecasting in irrigation systems. Weather forecasting can efficiently help in increasing crop productivity and optimizing irrigation systems. Weather forecasting can efficiently help in increasing crop productivity and optimizing irrigation systems. The integration of weather forecasting can efficiently help in increasing crop productivity and optimizing irrigation systems. Weather forecasting can efficiently help in increasing crop productivity and optimizing irrigation systems. The integration of weather forecasting can efficiently help in increasing crop productivity and optimizing irrigation systems.

Recently, machine learning techniques have been incorporated into smart irrigation systems. Techniques such as regression, support vector machine, and neural networks have been used for soil moisture level prediction. The use of these techniques has shown better accuracy in prediction compared to other techniques [3][5][9].

Advanced deep learning techniques such as recurrent neural networks (RNN) and long short-term memory (LSTM) have also been used for soil moisture prediction. These techniques have shown very high accuracy in prediction compared to other techniques. The use of these techniques has shown better efficiency in irrigation management [1].

Research in precision agriculture has shown that high-resolution sensor data plays a vital role in accurate soil moisture level prediction. IoT technology has helped in the development of such sensing systems. The data generated by these sensing systems can be effectively used for improving the efficiency of smart irrigation systems [3].

Some research has also shown the use of low-power techniques such as TinyML for efficient decision-making. The use of ESP32 has shown better efficiency in smart irrigation management. The use of such techniques has shown better efficiency in remote areas [5].

Although these techniques have shown better efficiency in smart irrigation management, most of these techniques use fixed threshold values. Moreover, some of these techniques require complex computational resources for efficient operation. Thus, there is a need for a simple, adaptive, and cost-effective smart irrigation system that can use multi-sensor data for efficient operation. The proposed work aims to develop such a smart irrigation system by integrating multi-sensor data, cloud-based monitoring, and soil behaviour modeling..

III. TERMINOLOGY

A. Internet of Things (IoT)

A device or a network of devices that are connected to the internet and are able to collect and send information.

B. ESP32

A small device that can connect to the WiFi network and collect information from sensors, sending the information to the cloud.

C. Soil Moisture

The amount of water that a particular area of soil contains. It is a key factor that helps in determining whether or not crops are dry.

D. Predictive Modelling

A technique that involves analyzing the information available and predicting what may happen in the future. It predicts when the soil will dry up, enabling early irrigation.

E. Precision Agriculture

A technique of farming that involves the efficient use of resources such as water, fertilizers, and electricity.

F. Cloud-Based Monitoring

A technique of monitoring crops and conditions by collecting information from devices and displaying it online.

IV. METHODOLOGY

The proposed system is an adaptive smart irrigation system that integrates multiple sensors, an ESP32 microcontroller, and a cloud-based platform. The system continuously monitors soil and environmental conditions and makes intelligent irrigation decisions based on predictive analysis.

The soil moisture sensor measures the water content in the soil, while the temperature and humidity sensors monitor environmental conditions. The ESP32 microcontroller processes this data and transmits it to the cloud. A relay module is used to control the water pump based on the irrigation decision.

The system continuously collects data from sensors and sends it to the ESP32. The microcontroller analyzes the data and uploads it to the cloud platform. Based on soil moisture levels and environmental conditions, the system predicts irrigation requirements. If the soil moisture falls below a threshold or is expected to drop rapidly, the system activates the pump; otherwise, it remains off.

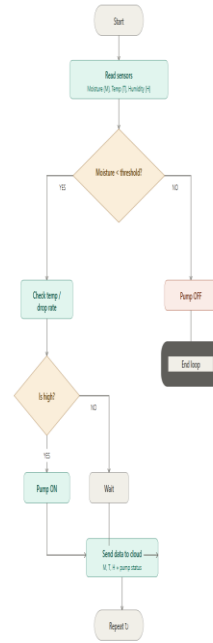


Fig. 1. System architecture of the proposed smart irrigation system.

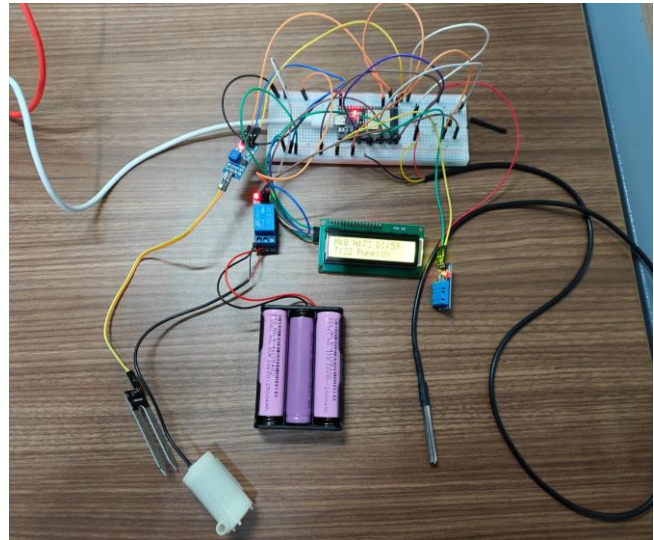


Fig. 2. Hardware setup of the smart irrigation system

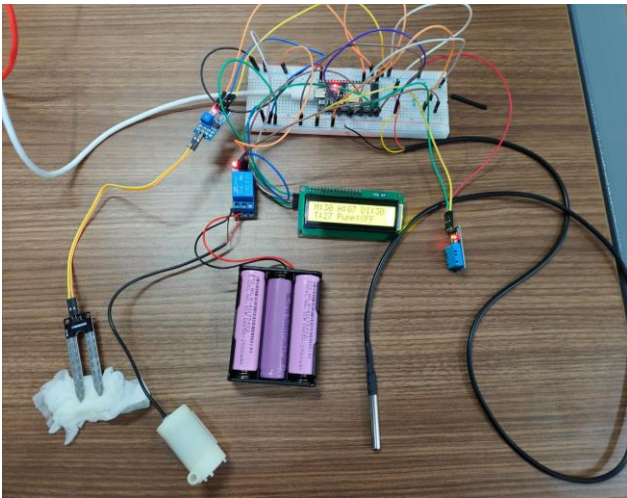


Fig. 3. System operation with moisture variation.

```

----- SENSOR DATA -----
Moisture (%): 0.00
Temperature (C): 22.31
Humidity (%): 73.30
Dryness Index: 59.32
Pump: ON
----- SENSOR DATA -----
Moisture (%): 0.00
Temperature (C): 22.31
Humidity (%): 73.30
Dryness Index: 59.32
Pump: ON

```

Fig. 4. Serial monitor output showing pump ON condition.

```

----- SENSOR DATA -----
Moisture (%): 30.00
Temperature (C): 27.37
Humidity (%): 87.30
Dryness Index: 29.91
Pump: OFF
----- SENSOR DATA -----
Moisture (%): 30.00
Temperature (C): 27.37
Humidity (%): 87.30
Dryness Index: 29.91
Pump: OFF

```

Fig. 5. Serial monitor output showing pump OFF condition.

```

1 #include <Wire.h>
2 #include <LiquidCrystal_I2C.h>
3 #include <OneWire.h>
4 #include <DallasTemperature.h>
5 #include <DHT.h>
6
7 // ----- PIN DEFINITIONS -----
8 #define MOISTURE_PIN 34
9 #define ONE_WIRE_BUS 4
10 #define RELAY_PIN 18
11 #define DHT_PIN 5
12
13 // ----- SENSOR TYPES -----
14 #define DHTTYPE DHT11 // or DHT22
15
16 // ----- LCD -----
17 LiquidCrystal_I2C lcd(0x27, 16, 2);
18
19 // ----- TEMP SENSOR -----
20 OneWire oneWire(ONE_WIRE_BUS);
21 DallasTemperature sensors(&oneWire);
22
23 // ----- HUMIDITY SENSOR -----
24 DHT dht(DHT_PIN, DHTTYPE);
25
26 // ----- PARAMETERS -----
27 float k_temp = 1.5;
28 float k_hum = 0.5; // humidity weight
29 float threshold = 50; // slightly higher now
30
31 void setup() {
32   Serial.begin(115200);
33
34   pinMode(RELAY_PIN, OUTPUT);
35   digitalWrite(RELAY_PIN, LOW); // OFF
36
37   lcd.init();
38   lcd.backlight();
39
40   sensors.begin();
41   dht.begin();
42

```

Fig. 6. Code for the hardware

```

42   Serial.println("==== Smart Irrigation System (Humidity Added) ===");
43 }
44
45 void loop() {
46   // ----- SOIL MOISTURE -----
47   int rawMoisture = analogRead(MOISTURE_PIN);
48   float moisture = map(rawMoisture, 4095, 0, 0, 100);
49
50   // ----- SOIL TEMPERATURE -----
51   sensors.requestTemperatures();
52   float temperature = sensors.getTempCByIndex(0);
53
54   // ----- HUMIDITY -----
55   float humidity = dht.readHumidity();
56
57   // ----- CHECK SENSOR ERROR -----
58   if (isnan(humidity)) {
59     Serial.println("DHT Error!");
60     humidity = 0;
61   }
62
63   // ----- IMPROVED DRYNESS INDEX -----
64   float DI = (100 - moisture)
65     + k_temp * (temperature - 25)
66     - k_hum * (humidity);
67
68   // ----- SERIAL OUTPUT -----
69   Serial.println("----- SENSOR DATA -----");
70
71   Serial.print("Moisture (%): ");
72   Serial.println(moisture);
73
74   Serial.print("Temperature (C): ");
75   Serial.println(temperature);
76
77   Serial.print("Humidity (%): ");
78   Serial.println(humidity);
79
80   Serial.print("Dryness Index: ");
81   Serial.println(DI);
82

```

Fig. 7. Continuation of the code

```

84   // ----- CONTROL LOGIC -----
85   if (DI > threshold) {
86     digitalWrite(RELAY_PIN, HIGH); // ON
87     Serial.println("Pump: ON");
88   } else {
89     digitalWrite(RELAY_PIN, LOW); // OFF
90     Serial.println("Pump: OFF");
91   }
92
93   // ----- LCD DISPLAY -----
94   lcd.clear();
95
96   // First Row
97   lcd.setCursor(0, 0);
98   lcd.print("M:");
99   lcd.print(moisture, 0);
100
101   lcd.print(" H:");
102   lcd.print(humidity, 0);
103
104   lcd.print(" DI:");
105   lcd.print(DI, 0);
106
107   // Second Row
108   lcd.setCursor(0, 1);
109   lcd.print("T:");
110   lcd.print(temperature, 0);
111
112   lcd.print(" ");
113
114   if (DI > threshold) {
115     lcd.print("Pump:ON");
116   } else {
117     lcd.print("Pump:OFF");
118   }
119
120 }

```

Fig. 8. Continuation of the code

V. RESULTS AND DISCUSSION

A. Dataset

The proposed project has data collected for five crops, comprising a combination of physically collected data and data obtained from the past experiences and reports .

B. Crop Analysis

1) Paddy (*Oryza sativa*): High water demand (45% threshold). Results show frequent irrigation trigger signals. The system interprets the saturation point perfectly, stopping irrigation at 55% to prevent deep percolation.

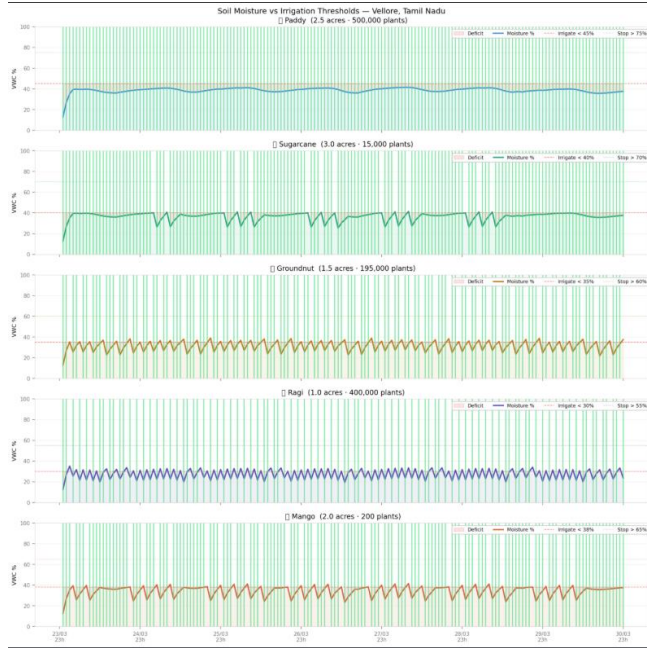


Fig. 9. Paddy moisture data and irrigation triggers.

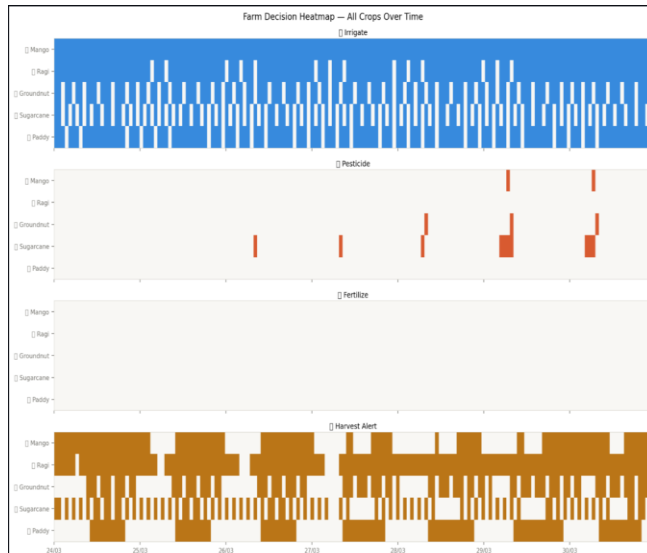


Fig. 10. Paddy temperature and humidity correlation.

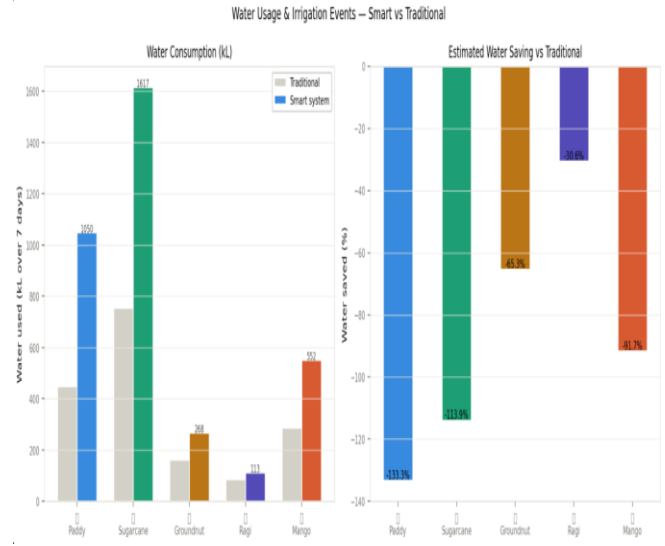


Fig. 11. Paddy irrigation event log.

2) Sugarcane (*Saccharum officinarum*): Long growth cycle (360 days), moderate moisture threshold (40%). The system interprets evapotranspiration (ET_0) data to predict irrigation timing before the plant wilts, resulting in more consistent stalk growth and higher sugar content.

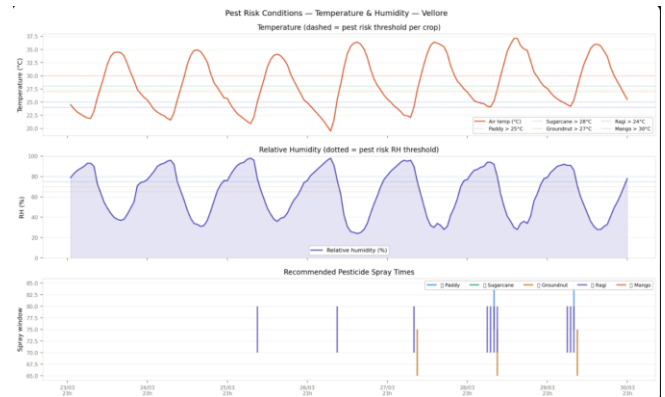


Fig. 12. Sugarcane moisture and ET_0 analysis.

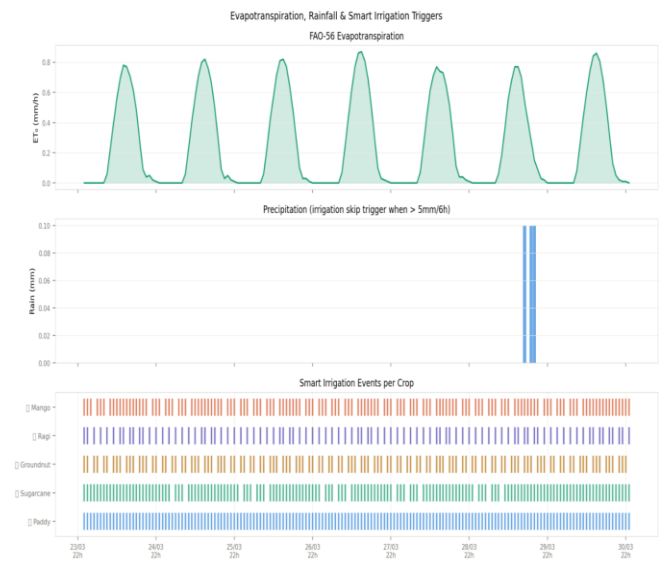


Fig. 13. Sugarcane irrigation schedule.

3) Groundnut (*Arachis hypogaea*): Lower moisture threshold (35%), high humidity sensitivity. The app flags pest risk when

temperature/RH signals indicate conditions favourable for fungal disease, reducing pesticide use.

4) Ragi (Finger Millet): Lowest moisture threshold (30%), extremely drought-resistant. Results often show idle status, demonstrating system efficiency through multi-crop logic that avoids wasting resources on hardy crops.

5) Mango (*Mangifera indica*): High temperature tolerance ($>30^{\circ}\text{C}$ threshold for risk). During the flowering stage, the system interprets the heat index to suggest micro-irrigation to cool the canopy and prevent flower drop.

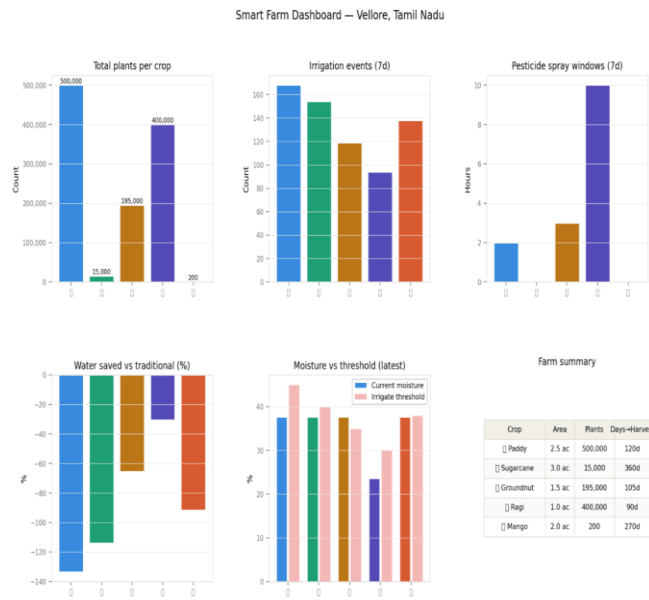


Fig. 14. Mango heat index and irrigation response.

VI. CONCLUSION

The proposed system for an adaptive multi-sensor predictive smart irrigation system effectively illustrates the application of IoT as well as soil behavior modeling for the efficient management of agricultural resources. The system includes the integration of various sensors such as soil moisture, temperature, humidity sensors, as well as an ESP32 microcontroller. The system is capable of monitoring the real-time environment through the data received from the sensors. The data is then used to make intelligent decisions regarding the irrigation process, enabling the automatic operation of the pump through the relay mechanism.

The inclusion of the cloud-based monitoring system with the help of ThingSpeak enhances the functionality of the system. The system is not only capable of monitoring the real-time data but also enables the storage of the data with the help of the cloud-based system. Unlike traditional irrigation systems, the proposed system includes the prediction of the soil moisture level.

The experimental results illustrate the efficiency of the system in the effective management of agricultural resources. The system not only reduces the wastage of resources but also helps maintain the optimal condition required for the growth of crops.

A. Key Findings

- Soil moisture sensor gave accurate information for irrigation decision-making in real time.

- Temperature and humidity are significant factors affecting depletion of soil moisture.
- Predictive model based on moisture drop rate was useful for anticipating irrigation requirements.
- Cloud monitoring enabled efficient data visualization and was effectively incorporated.
- Automation of irrigation was successfully carried out with improved efficiency.
- Proposed approach maintained optimal soil moisture levels consistently.

B. Future Work

- Integration of weather forecasting APIs to avoid irrigating during rainfall.
- Implementation of machine learning algorithms for better soil moisture estimation.
- Development of a mobile application for real-time monitoring and control.
- Extension to multi-zone irrigation systems for large agricultural fields.
- Integration of solar power for energy-efficient and sustainable operation.
- Integration of pH and nutrient sensors for comprehensive soil analysis.

C. Limitations

- Accuracy depends on sensor calibration; inaccurate readings could affect irrigation.
- Predictive model may not be accurate in all environmental conditions.
- System requires a stable internet connection for cloud platform access.
- Currently suitable for small-scale applications; scaling needed for large fields.
- Dependent on consistent power supply.

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